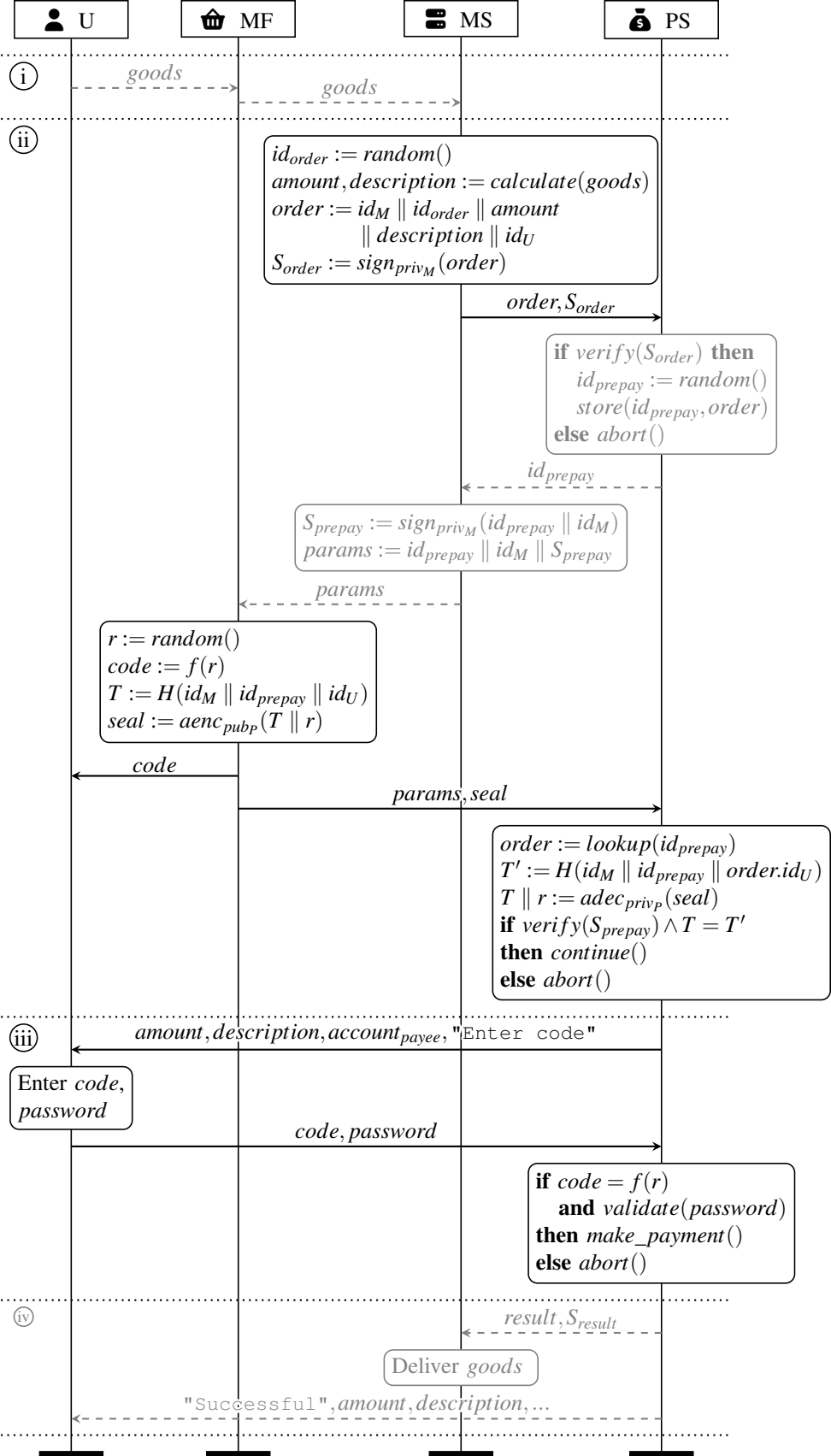

U
MF
MS
PS
i
goods
goods
ii
$id_{order} := random()$
$amount, description := calculate(goods)$
$order := id_M \parallel id_{order} \parallel amount \parallel description \parallel id_U$
$S_{order} := sign_{priv_M}(order)$
$order, S_{order}$
if $verify(S_{order})$ **then**
$id_{prepay} := random()$
$store(id_{prepay}, order)$
else $abort()$
id_{prepay}
$S_{prepay} := sign_{priv_M}(id_{prepay} \parallel id_M)$
$params := id_{prepay} \parallel id_M \parallel S_{prepay}$
$params$
$r := random()$
$code := f(r)$
$T := H(id_M \parallel id_{prepay} \parallel id_U)$
$seal := aenc_{pub_P}(T \parallel r)$
$code$
$params, seal$
$order := lookup(id_{prepay})$
$T' := H(id_M \parallel id_{prepay} \parallel order.id_U)$
$T \parallel r := adec_{priv_P}(seal)$
if $verify(S_{prepay}) \wedge T = T'$
then $continue()$
else $abort()$
iii
$amount, description, account_{payee}$, "Enter code"
Enter *code*, *password*
$code, password$
if $code = f(r)$
and $validate(password)$
then $make_payment()$
else $abort()$
iv
$result, S_{result}$
Deliver *goods*
"Successful", $amount, description, ...$